IE 7615: Neural Networks/Deep Learning

Generative Project Milestone – 2

GROUP 6

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GITHUB LINK : https://github.com/poornikajalavadi/Generative-Project-Neural-Networks?tab=readme-ov-file#1-setup-and-dependencies

Submission Date: 03/29/2026

ABSTRACT

The goal of this project is to develop an image captioning system by combining computer vision and natural language processing techniques. The model uses CLIP to extract meaningful visual features from images and GPT-2 to generate descriptive captions. A subset of the COCO dataset was used for training, where image-caption pairs were processed and converted into embeddings and tokenized sequences. In the second milestone, a model was built to connect visual embeddings with a language model for caption generation. The results show that the model is capable of producing relevant and grammatically correct captions, demonstrating the effectiveness of combining vision and language models for multimodal tasks.

INTRODUCTION

Image captioning is an important task that combines computer vision and natural language processing to generate textual descriptions for images. It has applications in areas such as accessibility, content search, and automated systems. In this project, we aim to build an image captioning model using a combination of CLIP and GPT-2. CLIP is used to extract meaningful visual features from images, while GPT-2 is used to generate captions based on these features. The project is divided into two milestones. The first milestone focuses on data preprocessing, embedding generation, and caption tokenization. The second milestone focuses on building and evaluating the model for caption generation. This approach helps in understanding how visual and textual representations can be combined effectively.

MILESTONE 1 SUMMARY

- A subset of 5000 images was selected from the COCO dataset, out of which 2811 valid image-caption pairs were used after filtering missing data.
- CLIP was used to extract 512-dimensional embeddings representing visual features of images.
- GPT-2 tokenizer was applied to convert captions into token sequences with special tokens for start, end, and padding.
- The processed data, including embeddings and tokenized captions, was saved for further use in model training.

MILESTONE-2 : MODEL DEVELOPMENT AND CAPTION GENERATION

MODEL ARCHITECTURE

The model is designed by combining two powerful components: a visual encoder and a language model. First, CLIP is used to understand the image. It converts each image into a numerical representation (embedding) that captures important visual features like objects, scenes, and context. These embeddings are of fixed size and act as a summary of the image.

However, GPT-2, which is a language model, cannot directly understand image embeddings. To solve this, a projection layer is introduced. This layer transforms the CLIP embeddings into a format that GPT-2 can process. In simple terms, it acts like a bridge between image understanding and text generation.

Once the embeddings are converted, they are fed into GPT-2 to generate captions word by word. Two strategies are used in this process. In prefix conditioning, the image information is added at the beginning of the text sequence so that GPT-2 uses it as context while generating captions. In cross-attention, the model directly looks at image features while generating each word, which allows it to produce more context-aware captions.

Overall, this architecture allows the model to connect visual understanding with natural language generation in a smooth and effective way.

TRAINING PROCESS

The training process focuses on teaching the model how to connect images with their correct captions. Instead of training from scratch, precomputed image embeddings from CLIP and tokenized captions from GPT-2 are used as inputs.

The model is trained in batches to efficiently handle large data and memory limitations. During training, the model tries to predict the next word in a caption based on the image features and previously generated words. If the prediction is incorrect, a loss function calculates the error, and the model updates its parameters to improve future predictions.

Training was performed on a GPU (Tesla T4), which significantly speeds up the computation. As training progresses, the loss gradually decreases, indicating that the model is learning the relationship between visual features and textual descriptions.

Overall, the training process helps the model improve its ability to generate meaningful and accurate captions for images.

CAPTION GENERATION

After training, the model is used to generate captions for new images. The image is first passed through CLIP to extract its embedding, which captures the visual meaning of the image. This embedding is then provided to GPT-2, which generates a caption one word at a time.

The generation process is sequential, meaning each word depends on the previous words and the image context. This helps the model produce grammatically correct and meaningful sentences. In many cases, the generated captions correctly describe the main objects and actions in the image.

However, some captions tend to be more general rather than highly detailed. This happens because the model is trained on limited data and sometimes focuses on common patterns instead of fine details.

Overall, the caption generation process demonstrates how visual and textual information can be combined to produce natural language descriptions from images.

DECODING STRATEGY

The caption generation process depends on the decoding strategy used by the language model. Instead of simply selecting the most probable next word every time, different strategies can be applied to improve the quality of generated captions.

In this project, decoding techniques such as greedy decoding and sampling were explored. Greedy decoding selects the most likely word at each step, which often produces stable but sometimes repetitive captions. On the other hand, sampling-based approaches introduce randomness, allowing the model to generate more diverse and natural-sounding captions.

Parameters like temperature can also be adjusted to control the randomness of the output. A lower temperature results in more deterministic captions, while a higher temperature produces more creative but less accurate results.

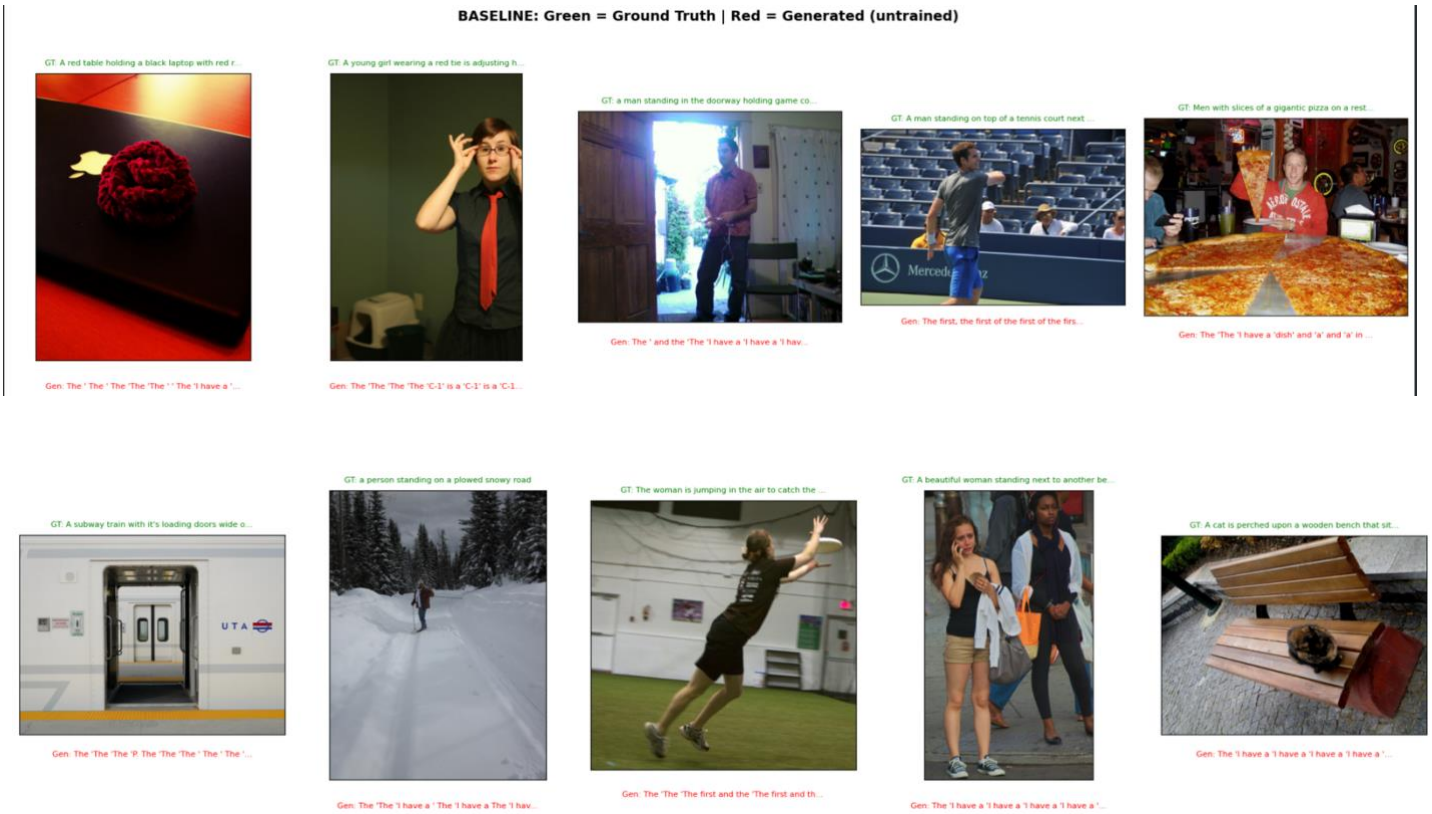
Overall, the choice of decoding strategy plays an important role in balancing accuracy and diversity in caption generation.

RESULTS AND OUTPUT ANALYSIS

The results of the model were evaluated by visualizing sample images along with their corresponding captions and similarity outputs. These results help in understanding how well the model captures visual features and generates meaningful textual descriptions. The following outputs demonstrate the performance of the model.

BEFORE TRAINING RESULT

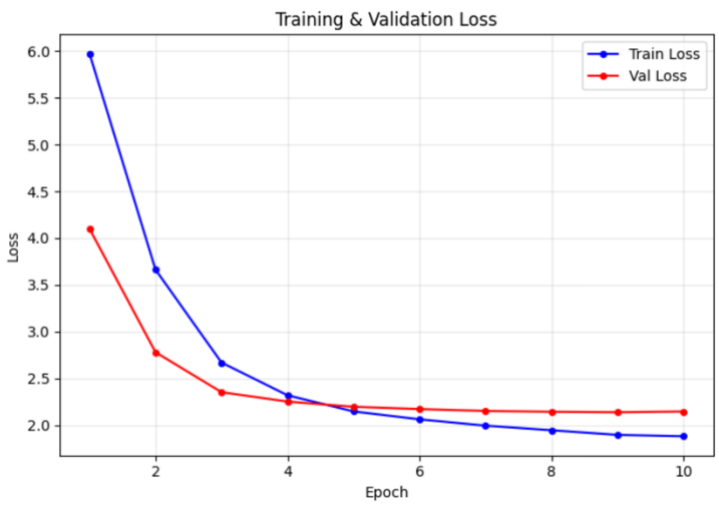
The figure shows the baseline results of the model before training. The generated captions (in red) are mostly incorrect and repetitive, indicating that the model has not yet learned the relationship between images and text. This highlights the importance of training in improving caption quality.

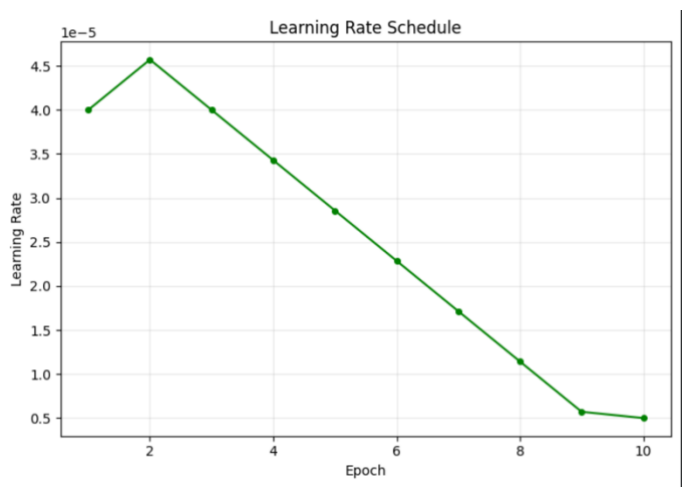


Baseline results before training showing poor caption generation compared to ground truth.

TRAINING PROCESS

The training process was monitored using training and validation loss curves. As shown in the figure, both training and validation loss decrease steadily over epochs, indicating that the model is learning effectively. The gap between training and validation loss remains small, which suggests that the model is not overfitting and generalizes well to unseen data. Additionally, a learning rate schedule was used, where the learning rate gradually decreases over time. This helps in stabilizing the training process and allows the model to converge more smoothly.

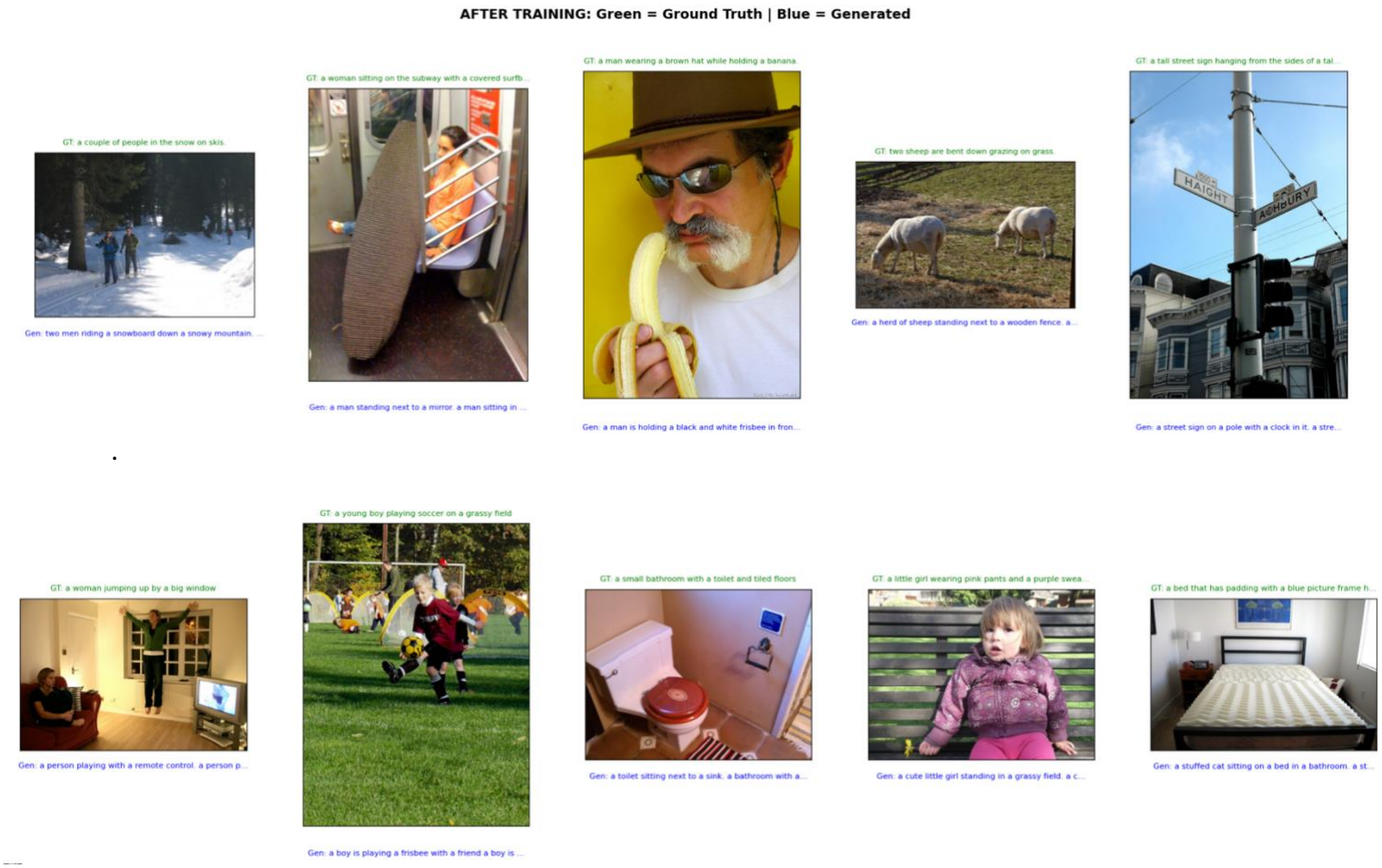




Training and validation loss curves along with learning rate schedule showing stable and effective model training.

AFTER TRAINING RESULT

The figure shows the results after training the model. Compared to the baseline results, the generated captions (in blue) are significantly more meaningful and relevant to the images. The model is able to correctly identify key objects and actions in many cases. Although some captions are not perfectly accurate, they still capture the general context of the image. This demonstrates that the model has successfully learned the relationship between visual features and textual descriptions



Caption generation results after training showing improved alignment with ground truth descriptions.

DECODING STRATEGY COMPARISON

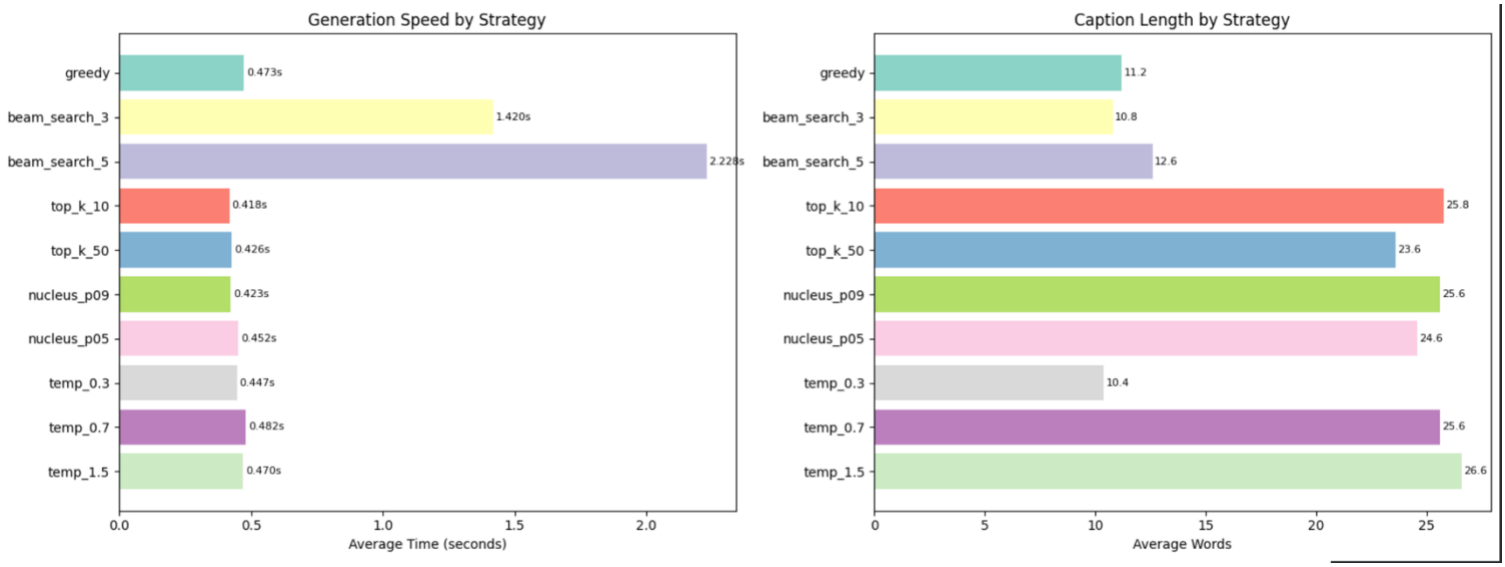
The figure compares different decoding strategies in terms of generation speed and caption length. It can be observed that greedy decoding is the fastest method but produces shorter and less detailed captions. Beam search generates more structured captions but is significantly slower. Sampling-based methods such as top-k and nucleus sampling provide a balance by generating longer and more descriptive captions while maintaining reasonable speed. Temperature-based decoding also influences caption diversity, where higher temperature results in longer and more varied captions.

Generation Speed (left)

- Greedy, top-k, nucleus -- fast (~0.4–0.5s)
- Beam search -- slow (up to ~2.2s)

Caption Length (right)

- Greedy -- shorter captions
- Sampling (top-k, nucleus, temperature) -- longer, more descriptive captions

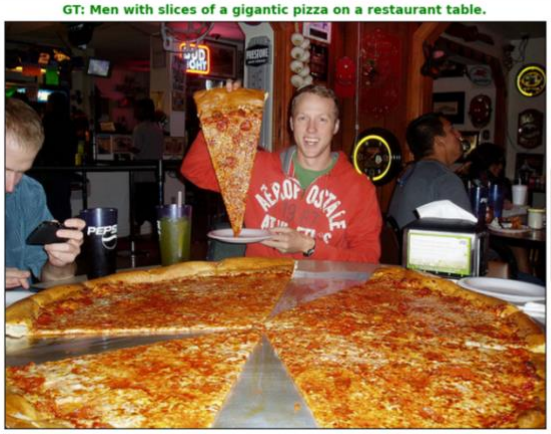


Comparison of different decoding strategies based on generation speed and caption length

CAPTION GENERATION RESULTS

The following examples demonstrate how different decoding strategies generate captions for the same image. It can be observed that the quality, coherence, and diversity of captions vary depending on the decoding method used.

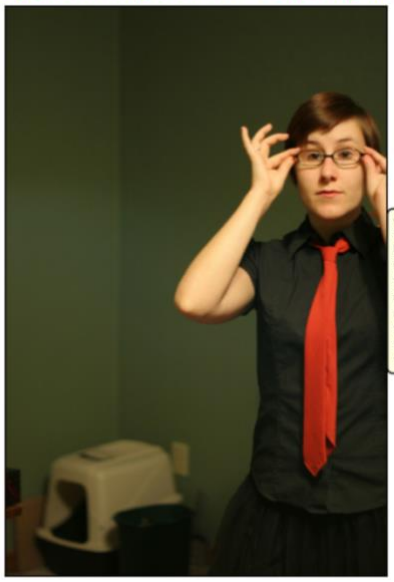
GT: Men with slices of a gigantic pizza on a restaurant table.



```
greedy: The first time the new "I have a new, and the firs...
beam_search_3: " " "
beam_search_5: " " " "
top_k_10: .. " the the way of the government has been in the...
top_k_50: -. It was, the's for the 'G. Al. The new "Ghee in ...
nucleus_p09: Capt. Harlan R. A. Having told me the important li...
nucleus_p05: . " ' … a " and The U.S.S.S. state of the U.S.S. s...
temp_0.3: The "The first time I have a problem with the "I d...
temp_0.7: ,-- and, -. and the only last I could be the most ...
temp_1.5: regardless hypocrisy comedian OPOOOO Doubt the 8. ...
```

Greedy decoding gives generic and incomplete captions, whereas beam search improves structure slightly. Sampling techniques produce more detailed descriptions but may include irrelevant content. This shows a trade-off between diversity and accuracy.

GT: A young girl wearing a red tie is adjusting her glasses.



```
greedy: The first,
beam_search_3: The first, "
beam_search_5: The first, "
top_k_10: , a "- and and and the one and. I'm not the one th...
top_k_50: ('s. . or: T' and's that, and the people, the worl...
nucleus_p09: The 40- (He is, produce the sum of of in a matter,...
nucleus_p05: - with the J, it is a. and for the whole of the fi...
temp_0.3: . The first, The "
temp_0.7: 1- and in a a " -, but the government of the team ...
temp_1.5: sheets their -Next part A ground calling the oblig...
```

Greedy decoding produces short and repetitive captions, while beam search gives slightly better structure. Sampling methods generate more diverse captions but sometimes lose coherence. Overall, better strategies improve variety but not always accuracy.

Caption outputs generated using multiple decoding strategies for a sample image.

KEY FINDINGS

- The model improves caption quality after training compared to initial outputs.
- Greedy decoding is fast but produces repetitive and less meaningful captions.
- Beam search generates more structured but limited captions.
- Sampling methods (top-k, nucleus) provide more diverse and descriptive outputs.
- There is a clear trade-off between diversity and coherence in generated captions.
- Overall, the model captures basic objects and scenes but struggles with detailed understanding.

CONCLUSION

The project successfully demonstrates an image captioning pipeline using deep learning techniques. The model is able to generate basic descriptions of images, and its performance improves with better decoding strategies. While advanced methods increase diversity, maintaining accuracy remains a challenge. This work highlights the importance of balancing efficiency, coherence, and richness in caption generation.

FUTURE IMPROVEMENTS

- Use a larger and more diverse dataset for better generalization
- Improve model architecture using attention mechanisms or transformers
- Fine-tune hyperparameters for better performance
- Apply evaluation metrics like BLEU score for quantitative analysis
- Reduce repetitive captions using better decoding techniques